

Information Retrieval: Assignment 2

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In this assignment, I implemented Ukrainian Creative Writing Agent. The agent has below functions:

- Stylistic transformation: Based on given input for a certain criteria (Transformation Direction, Modern Sub-genre (From), Classical Author (To)), the agent makes stylistic transformation for given text or example poetry. Each criteria has its option for select as below:
 - Transformation Direction: Modern → Classical, Modern → Classical
 - Modern Sub-genre (From): Internet Slang / Gen Z, Ukrainian Cyberpunk / Sci-Fi, 1980s Kyivan Underground, Lviv Hvara (Batiar Slang)
 - Classical Author (To): Taras Shevchenko (Romanticism), Ivan Franko (Realism), Lesya Ukrainka (Neoromanticism), Ivan Kotliarevsky (Burlesque), Vasyl Stus (Dissident Poetry)
- Phonetic Flow & Multi-Style Polishing: Based on input for Polishing Register / Genre, the agent review the given text and revise it so that the given text can correspond to the given genre:
 - Polishing Register / Genre: Poetry (Metrical & Rhyme Analysis), Business & Official Prose (Euphony & Terminology Polish), Everyday Dialogue & Conversation (Natural Flow & Ease)

Besides, the agent return Text-To-Speech for output text in Ukrainian with real-time Karaoke-style subtitle highlights so that users could learn how to read Ukrainian text appropriately and improve their Ukrainian pronunciation.

Also, heartbeat module and memory are introduced for the agent. With regard to memory, the agent compiles a brief entry including the target style and a snippet of the first line on generation completion, then normalizes and stores this in local storage. There is restriction for entries to the last two generations (capped at max 450 characters / 90 tokens) to maintain perfect style continuity at near-zero token cost.

Github link¹, HuggingFace working demo², and video link for demo³ are mentioned in footnote in this page.

In this assignment, compared to previous assignment, I did much more try and error to improve my AI agent performance, make UI more user-friendly design one, add or

¹https://github.com/katsufu/UUIR-assignment2_aiagent.git

²<https://huggingface.co/spaces/katsufu/ukrainian-writing-agent>

³<https://youtu.be/kt8ICAD9DWI>

integrate the agent's skills, etc. Especially, due to the structure of AI agent based on several modules in TypeScript language, it was not intuitively understandable for me to fix or check bugs, and grasp what is happening in each module and how they are related with each other. I did create visualization for my understanding about it, using Google Antigravity. As a result, it was great opportunity for me to create own AI agent and, moreover, to come up with idea for creating the agent, which is actually challenging in real world.